

A merger-induced starburst in the ancient Milky Way

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ABSTRACT

We use accurate estimates

Key words: stars: kinematics and dynamics – Galaxy: evolution – Galaxy: formation – Galaxy: abundances – Galaxy: stellar content – Galaxy: structure

1 INTRODUCTION

Mergers between galaxies are a sure way to produce a starburst provided there is gas available, either brought-in, native or both, to convert into stars. Across the spectrum of galaxy mass-ratios, from major (similar masses) and minor (accretion of a smaller object) mergers, interactions usually lead to enhanced star-formation activity (see e.g. Barnes & Hernquist 1991; Mihos & Hernquist 1994, 1996; Springel 2000; Cox et al. 2008; Di Matteo et al. 2008; Renaud et al. 2014). These bursts are however short-lived

2 DATA AND SAMPLE SELECTION

We use data from the APOGEE Data Release 17 (ADR17, Abdurro’uf et al. 2021). The chemical abundance data is taken from the `allStarLite` catalogue available on the survey’s website. The stellar Galacto-centric coordinates, velocities and integrals of motion as well as their uncertainties are taken from the `AstroNN` value-added catalogue (see Leung & Bovy 2019; Mackereth & Bovy 2018) which assumes the `MWPotential2014` gravitational potential from Bovy (2015). The above catalogues take advantage of the *Gaia* Early Data Release 3 photometry and astrometry (Gaia Collaboration et al. 2021; Fabricius et al. 2021; Riello et al. 2021; Lindegren et al. 2021). We use the same selection procedure as outlined in Belokurov & Kravtsov (2022) in terms of cleaning the ADR17 data of duplicates and unreliable measurements, removing stars in and around known Galactic satellites, and restricting the sample to red giant stars with high quality kinematics and chemistry. As a precaution we limit our stars to heliocentric distance < 15 kpc.

Figure 1 gives the overview of the stellar properties in our base ADR17 sample. Left panel of the Figure shows the distribution of the stars in the plane of total energy E and the vertical component of the angular momentum L_z . Most of the stars in the sample are in the disk and hence pile up along the line corresponding to the value of maximal L_z at given energy (right hand-side of the E - L_z plane in our chosen notation). A gap in the density distribution around $E^{-5} \approx -0.75$ is due to the particular shape

of the filed-of-view of the APOGEE survey. Middle panel of the Figure gives the logarithm of the stellar density in the plane of $[\text{Mg}/\text{Fe}]$ and $[\text{Fe}/\text{H}]$. Simple cuts are used to carve out three distinct populations: i) the accreted stars, predominantly belonging to the GS/E merger; these show the fastest decline in $[\text{Mg}/\text{Fe}]$ starting at around $[\text{Fe}/\text{H}] \approx -1.3$; ii) the high- α population corresponding to the Milky Way’s early disk as well as the Galaxy’s pre-disk phase, known as *Aurora* (see Belokurov & Kravtsov 2022), and finally, iii) the low- α population corresponding to the late stage of the disk formation, that following the interaction with the GS/E progenitor. Note that while at high metallicities, i.e. $[\text{Fe}/\text{H}] > -0.5$, $[\text{Mg}/\text{Fe}]$ ratio in this third group is at the lowest level, i.e. $[\text{Mg}/\text{Fe}] < 0.1$, the sequence is actually intermediate in terms of its α abundances at lower metallicities, i.e. $[\text{Fe}/\text{H}] < -1$ where it sits between the accreted and the high- α stars. Finally, the right panel of Figure 1 shows the stellar density in the plane of $[\text{Al}/\text{Fe}]$ and $[\text{Fe}/\text{H}]$. Three distinct populations can be readily identified here: the accreted halo dominate by the GS/E debris with low values of $[\text{Al}/\text{Fe}]$, not exceeding $[\text{Al}/\text{Fe}] \approx -0.1$ and the high- and low-alpha disks, with $[\text{Al}/\text{Fe}]$ values typically above -0.1 . Note that at $[\text{Fe}/\text{H}] < -1$, the in-situ stars reside in non-rotating pre-disk population known as *Aurora* (see Belokurov & Kravtsov 2022).

Figure 2 highlights differences in behaviour of $[\text{Al}/\text{Fe}]$ as a function of metallicity for the three stellar populations selected in the middle panel of Figure 1. Note that Figure 2 gives column-normalised density for each sample of stars. Moreover, we limit the accreted population (left) to have $|L_z| < 500$ km s⁻¹. No additional cuts (apart from those in $[\text{Mg}/\$

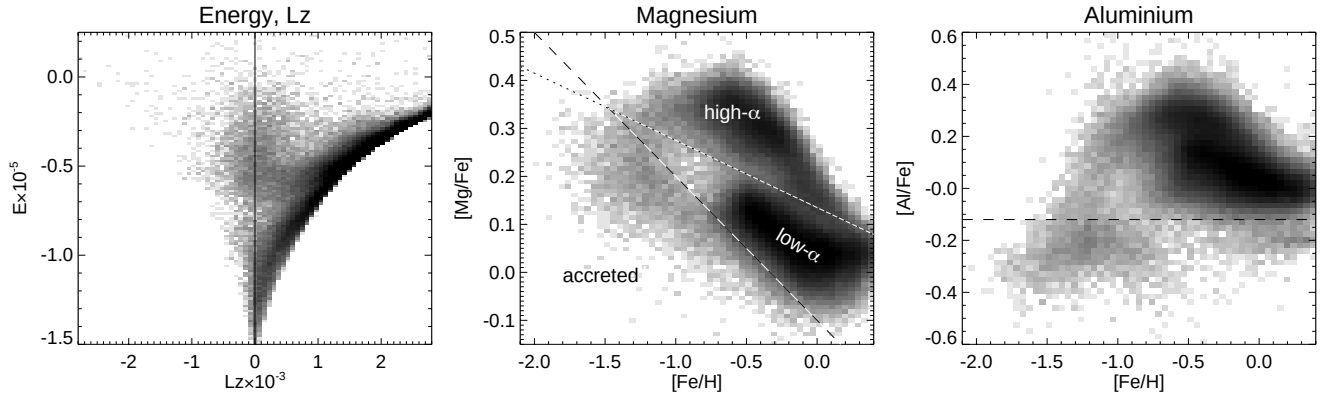


Figure 1. Orbital and chemical properties of the base ADR17 sample (see Section 2 for details). *Left:* Logarithm of stellar density in the space of total energy, E , and the vertical component of the angular momentum, L_z . Solid vertical line corresponds to $L_z = 0$. *Middle:* Logarithm of stellar density in the space of $[Fe/H]$ and $[Mg/Fe]$. Diagonal straight lines show the selection of the accreted, high- α and low- α populations. *Right:* Logarithm of stellar density in the space of $[Fe/H]$ and $[Al/Fe]$. Horizontal dashed line demarcates the in-situ (higher $[Al/Fe]$) and the accreted (lower $[Al/Fe]$) stars.

past $[Al/Fe] \approx -0.2$. This nearly flat pattern can be contrasted to the $[Al/Fe]$ sequence of the high- α population (middle) which rises quickly above $[Al/Fe] \approx -0.2$ and reaches $[Al/Fe] \approx -0.3$ around $[Fe/H] \approx -1$.

The low- α sequence displays a behaviour distinct from both the accreted and the high- α stars. First, the rise in $[Al/Fe]$ is delayed, the sequence crosses $[Al/Fe] = -0.2$ mark around $[Fe/H] \approx -1$. Second, the $[Al/Fe]$ increase with metallicity is slower than that for the high- α stars. We use the $[Al/Fe]$ trends exhibited by three populations to build a clean sample of low- α stars. Below $[Al/Fe] = -0.12$ it is not possible to cleanly separate the accreted and the in-situ populations, therefore in the analysis below we focus on the in-situ stars with $[Al/Fe] > -0.12$. Also note that the high- and low-alpha sequences come close together at $-1.2 < [Fe/H] < -0.7$. To minimise the contamination of the low-alpha sequence by high-alpha stars, we apply an additional cut $[Al/Fe] > 0.9[Fe/H] + 0.9$ shown as a dashed diagonal line in the Figure. Note that typical low- α (sometimes referred to as thin disk) stellar samples are limited to $[Fe/H] > -0.6$. Here we extend this population by some 0.4 dex to $[Fe/H] \approx -1$.

Inspired by Figures 1 and 2 we identify stars belonging to the accreted, high- and low-alpha populations relying on chemistry only, i.e. with cuts in $[Mg/Fe]$, $[Al/Fe]$ and $[Fe/H]$ dimensions. Figure 3 shows the orbital properties of these three populations selected chemically. The accreted stars (left) display little rotation and are limited to $|L_z| < 0.5 \times 10^3$ but reach the highest values of total energy and (in this sample) are mostly limited to $E \times 10^{-5} > -0.6$. The high- α sample contains both the early (thick) disk of the Galaxy as well the in-situ halo which has a wide range of the angular momentum. Please note the existence of a population of high- α stars stretching from negative L_z (retrograde motion) to high positive L_z . These high- α stars on halo-like orbits belong to the Aurora (Belokurov & Kravtsov 2022) and Splash populations (Belokurov et al. 2020). Finally, the third panel of Figure 3 gives the orbital properties of the low- α sample. As expected, most of the low- α stars are found in a fast rotating (thin) disk of the Milky Way. Note however the presence of a small but noticeable group of stars with zero angular momentum around $E \times 10^{-5} \approx -0.55$. These low- α stars with halo-like orbits are distinct from both the accreted halo (because they occupy a different range of energies) and from the in-situ halo (because they have a

much narrower distribution of L_z , completely disconnected from that of the disk). We dub this population *Aura*.

3 LOW- α DISK STARS WITH HALO-LIKE ORBITS

Above we describe a simple set of selection cuts which yields a highly pure sample of stars dominated by the Galactic low- α (or thin) disk. Surprisingly, a small fraction of the stars in this group appears to have mean zero angular momentum, thus exhibiting halo-like orbital behaviour. Here we investigate further this

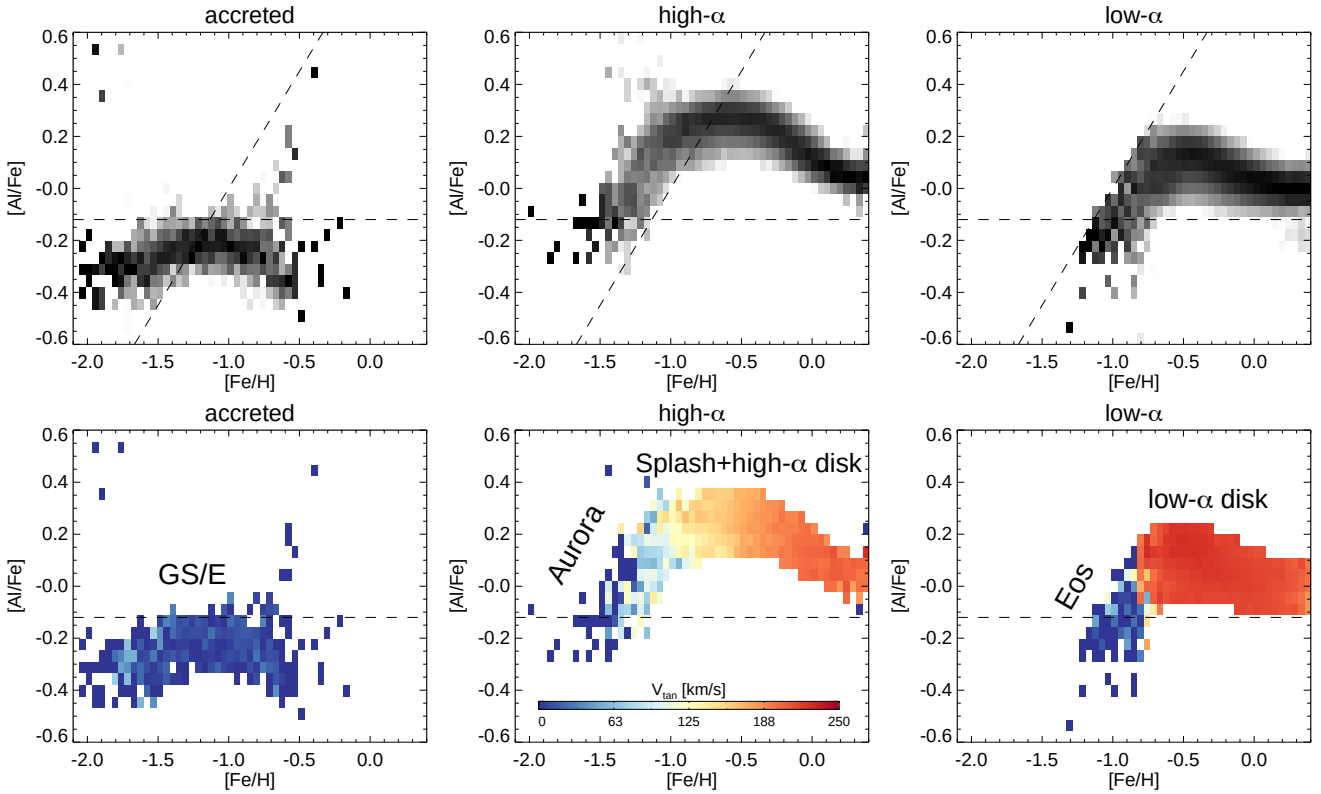


Figure 2. Aluminium abundance as a function of metallicity for the three populations selected using $[\text{Mg}/\text{Fe}]$ as shown in Figure 1. *Left:* Column-normalised density of the accreted stars (additional $|L_z| < 500$ cut was applied). Note that the bulk of this population is at $[\text{Al}/\text{Fe}] < -0.2$. *Middle:* Column-normalised density of the high- α sample. Note that at $-1.5 < [\text{Fe}/\text{H}] < -1$ this sample is dominated by the non-rotating, pre-disk Aurora component of the Galaxy (Belokurov & Kravtsov 2022). *Right:* Column-normalised density of the low- α stars. Horizontal dashed line separates the in-situ and the accreted stars. Diagonal dashed line separated the high- and the low- α stars.

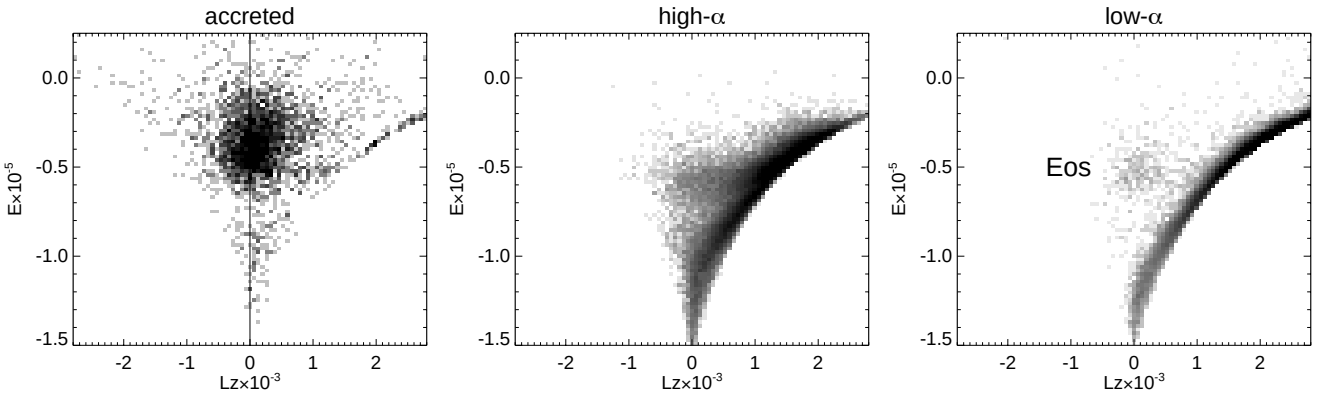


Figure 3. Orbital properties of the three chemically selected populations shown in Figure 2. *Left:* Logarithm of stellar density of the accreted stars in the space of E and L_z . *Middle:* Same as the previous panel but for the high- α stars. *Right:* Same as the two previous panels but for the low- α stars. Note the presence of the small population of stars with $|L_z| \approx 0$ at around $E \times 10^{-5} \approx -0.55$.

main cloud of stars with $[\text{Fe}/\text{H}] > -0.7$ and $V_{\text{tan}} > 180 \text{ km s}^{-1}$ - this is the thin disk, exhibits a correlation between the azimuthal velocity and the apo-centre: the higher V_{tan} values achieved by the stars whose apo-centres lie beyond the Solar neighborhood. This correlation may perhaps be understood considering the epicyclic motion of the stars on orbits with moderate eccentricity entering the Solar neighborhood. The corresponding distribution of peri-centric distances is given in the middle panel. Peri-centres of the rotating low- α population are typically between 4 and 12 kpc. At high

metallicity, i.e. $[\text{Fe}/\text{H}] > 0$, note a considerable population with low and negative values of V_{tan} . These mostly super-Solar metallicity stars have peri-centres of order of 1 kpc and apo-centric distances below 5-6 kpc. In other words, these stars belong to the bulge/bar region of the Galaxy with low rotation speeds and non-circular motions. In the right panel of the Figure the stars are colour-coded according to their age. Only measurements with relative age uncertainties lower than 20% are included. Note that stars on orbits with small spin are preferentially old.

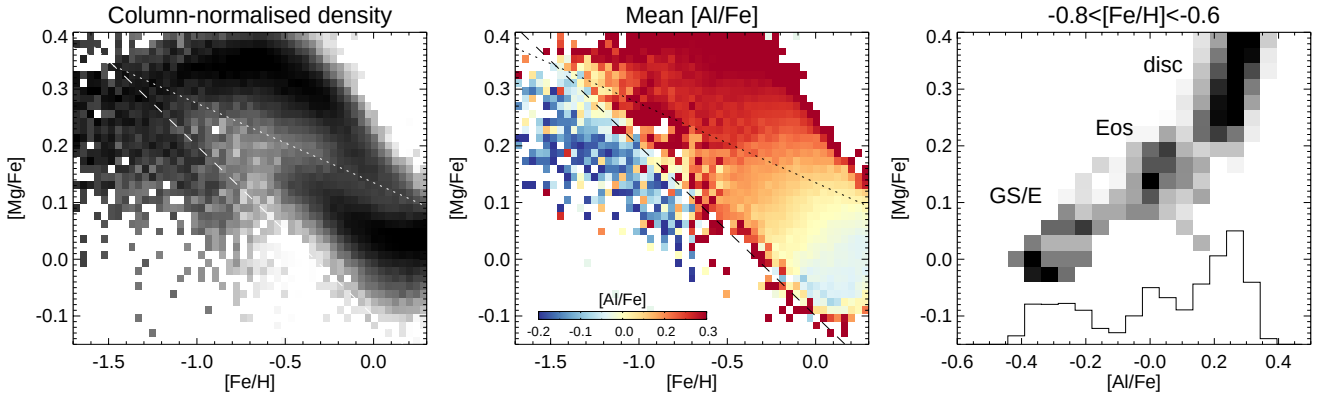


Figure 4. Three distinct populations in $[\text{Fe}/\text{H}]$ - $[\text{Mg}/\text{Fe}]$ - $[\text{Al}/\text{Fe}]$ space. *Left:* Column-normalised stellar density in $[\text{Mg}/\text{Fe}]$ vs $[\text{Fe}/\text{H}]$ space. Diagonal lines separating the accreted, high- α and low- α populations are the same as in Figure 1. *Middle:* The space shown in the previous panel now colour-coded with the mean $[\text{Al}/\text{Fe}]$ abundance ratio. Note that the dashed line separates stars with low (blue) and high (yellow/red) $[\text{Al}/\text{Fe}]$. *Right:* A slice through the 3D $[\text{Fe}/\text{H}]$ - $[\text{Mg}/\text{Fe}]$ - $[\text{Al}/\text{Fe}]$ space at constant metallicity $-0.8 < [\text{Fe}/\text{H}] < -0.6$. Three distinct groups of stars are clearly seen in this elongated density distribution. The black histogram shows the 1D version of the 2D density collapsed onto the $[\text{Al}/\text{Fe}]$ dimension.

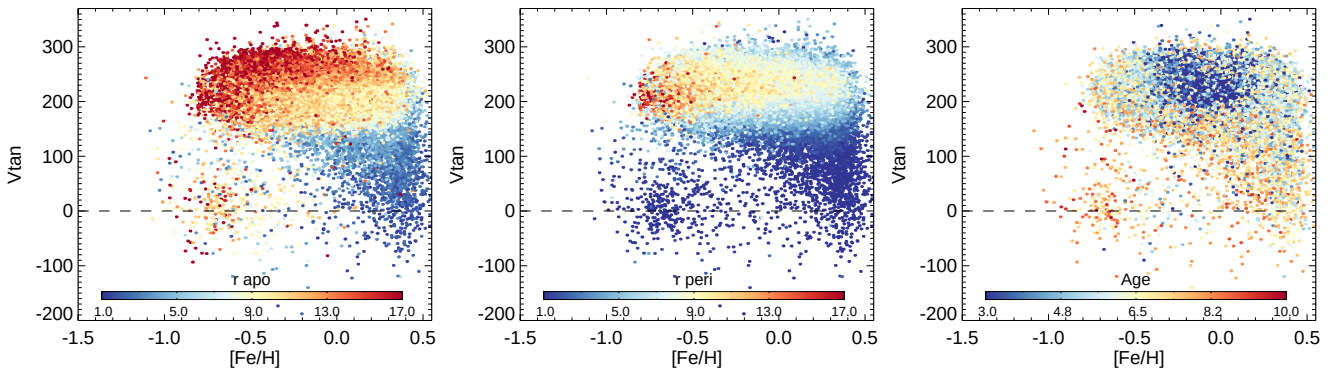


Figure 5. Azimuthal velocity V_{tan} vs metallicity $[\text{Fe}/\text{H}]$ for the low- α sample. There are three distinct groups of stars: the thin disk with high V_{tan} across a wide range of $[\text{Fe}/\text{H}]$, the bulge/bar with $[\text{Fe}/\text{H}] > 0$ and intermediate and low V_{tan} , and, finally, *Aura* with $[\text{Fe}/\text{H}] < -0.5$ and $V_{\text{tan}} \approx 0 \text{ km s}^{-1}$. *Left:* Stars are colour-coded according to their apo-centric distance. The thin disk stars have apo-centres larger than 8 kpc, the bulge/bar stars have apo-centres lower than 5 kpc, while the *Aura* stars have apo-centres similar to the thin disk, i.e. > 8 kpc. *Middle:* Colour-coding is according to the stellar pericentric distances. Stars with high V_{tan} are on nearly circular orbits (peri-centres and apo-centres are similar). The bulge/bar and *Aura* stars have very low peri-centres, namely of order of 1 kpc. *Right:* Age colour-coding. The thin disk is relatively young compared to both the bulge/bar and the *Aura* populations.

Figure 5 demonstrates that *Aura* is clearly different from the rest of the low- α population. At small (net zero) V_{tan} , *Aura* is distinct from the stars with $[\text{Fe}/\text{H}] > 0$ belonging to the bulge/bar of the Milky Way: the stars in *Aura* with $[\text{Fe}/\text{H}] < -0.5$ have apo-centres that are large, these orbits often straddle (and go beyond) the Solar neighborhood. *Aura* apo-centric distances are greater than 8-10 kpc but come as close as 1 kpc to the Galactic Centre, yielding eccentricities in the range of $0.8 < e < 1$, the highest amongst the low- α population. In Figure 5 not only there is a paucity of stars at fixed low azimuthal velocity between *Aura* and the bulge/bar, the *Aura* population appears to be well separated from the rotating low- α disk too. At the same metallicity $-1 < [\text{Fe}/\text{H}] < -0.5$ there is a clear gap in the V_{tan} distribution between *Aura* and the low- α disk. As shown in the right panel of the Figure, ages of the *Aura* stars are uniformly old, i.e. between 7 and 10 Gyr, as old or older compared to the oldest stars in the thin disk where the bulk of star was formed less than 5 Gyr ago. *Aura* is also different from the bulge/bar stars whose mix of ages is much broader.

Figure 6 investigates the age make-up of *Aura* further. The left panel shows the age-metallicity tracks of the high- α (blue)

and the low- α (red) populations. As above, only stars with relative age uncertainties lower than 20% are included. Not many high-quality ages above 10 Gyr can be found in the AstroNN catalogue (see Leung & Bovy 2019; Mackereth & Bovy 2018) in contrast with the age determinations using MS turn-off stars (see Haywood et al. 2013; Bonaca et al. 2020; Xiang & Rix 2022). This is largely due to the different systematic biases incurred by these two age determination methods. Additionally, the training set in the AstroNN catalogue does not contain many examples

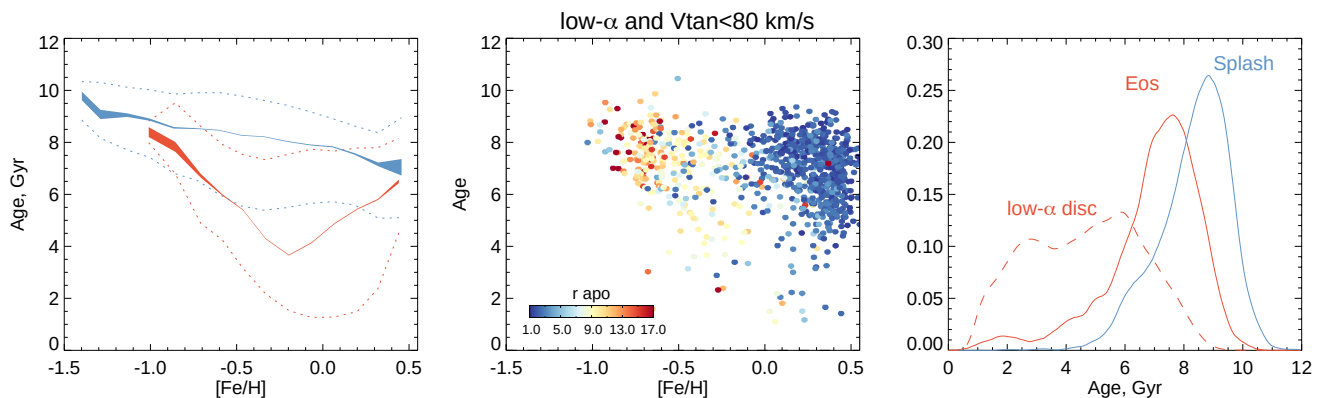


Figure 6. Age-metallicity behaviour of the high- α and the low- α populations. Only stars with relative age errors lower than 20% are included. *Left:* Median age at the fixed metallicity is shown for the high- α (blue) and low- α (red) samples. Dotted lines give the 3.3σ extent of the age distribution at given [Fe/H]. The high- α age-metallicity track is shallow, indicating a very fast enrichment. The equivalent track for the low- α population is much steeper out to [Fe/H] = -0.2 indicating a slower SF activity. Also, note a bent upwards at higher metallicities, due to the contribution of the stars from the inner Galaxy. *Middle:* Age-metallicity for stars with low azimuthal velocities, i.e. $V_{\text{tan}} < 80 \text{ km s}^{-1}$. Data-points are colour-coded according to the stars’ apo-centric distances. The metal-rich stars are mostly blue, indicating apocentres within $\approx 4 \text{ kpc}$ from the Galactic Centre. The *Aura* stars have much larger apo-centres, typically above 10 kpc and are predominantly old, with ages between 6 and 10 Gyr . The bulge/bar age distribution is wider, spanning a range from 4 to 8 Gyr . *Right:* Age distributions of the low- α disk (red dashed), *Aura* (red solid) and *Splash* (blue solid). Note a close-match between the *Splash* and the *Aura* age distributions.

in Figure 5, the older age stars at super-solar metallicities are predominantly bulge/bar denizens with apo-centres $< 5 \text{ kpc}$. This inverted Λ shape in the age-metallicity space has been seen before (see Feuillet et al. 2019; Xiang & Rix 2022).

The dominance at high metallicity of the stars residing in the bulge/bar region is illustrated in the middle panel of Figure 6 where only objects with $V_{\text{tan}} < 80 \text{ km s}^{-1}$ are shown, color-coded according to their apo-centric distances. Here two populations are clearly discernible: the *Aura* with [Fe/H] < -0.5 , apo-centres typically beyond 10 kpc and a narrow age distribution peaking around 8 Gyr , and the bulge/bar stars with [Fe/H] > 0 , small apo-centres ($< 4 \text{ kpc}$) and a broader age distribution. The right panel of the Figure compares the 1D age distributions of the low- α disk (dashed red), the *Aura* (solid red) and the *Splash* (solid blue). The *Splash* stars are identified here as those residing in the high- α sequence, with [Fe/H] > -0.9 and $V_{\text{tan}} < 80 \text{ km s}^{-1}$. The age distribution of the low- α disk is broad and extends from 1 to 8 Gyr back in time. The age distribution of the *Splash* stars is peaked around 9 Gyr . These stars formed in the high- α disk of the Galaxy and were later kicked out during the interaction between the MW and the GS/E progenitor (REF).

4 DISCUSSION AND CONCLUSIONS

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DATA AVAILABILITY

This study uses allStarLite-dr17-synspec_rev1 and apogee_astroNN-DR17 catalogues publicly available at https://www.sdss.org/dr17/irspec/spectro_data/.